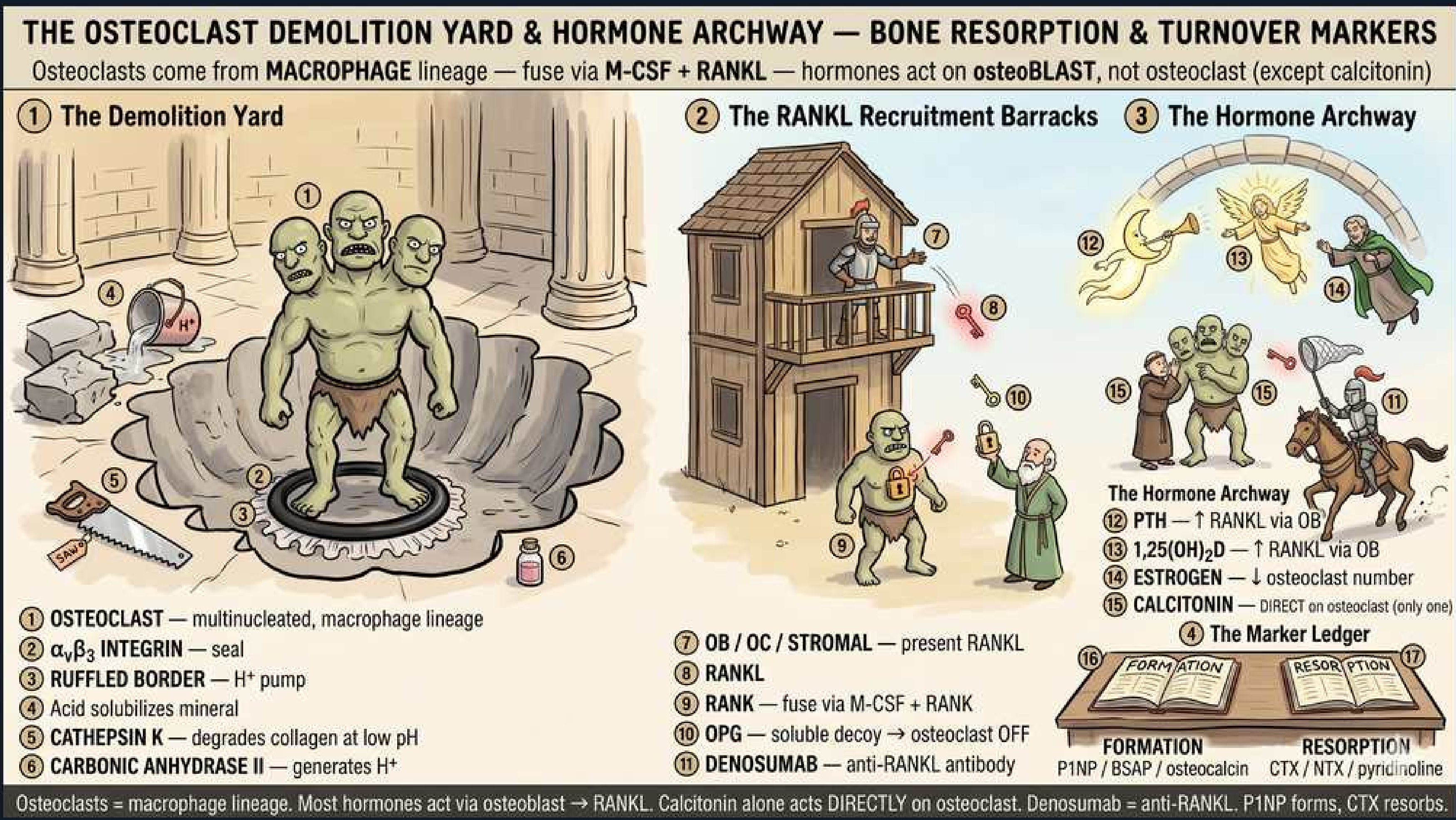
Bone Resorption & Turnover Markers — Osteoclast Machinery, RANKL/OPG, Hormones



1. Osteoclast — macrophage lineage

WHAT YOU SEE
Multinucleated green ogre in pit

WHAT IT ENCODES
Osteoclasts are large MULTINUCLEATED cells from the MACROPHAGE/monocyte (hematopoietic) lineage, formed by fusion driven by M-CSF + RANKL. They resorb bone in Howship's lacunae.

BOARD TRAP
Do NOT call osteoclasts mesenchymal — that's osteoblasts. The two cells share NO common precursor.

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2. αvβ3 integrin — seal

WHAT YOU SEE
Sealing ring at cell base

WHAT IT ENCODES
The alpha-v-beta-3 integrin anchors the osteoclast to bone and forms the sealing zone, creating an isolated resorption compartment beneath the ruffled border.

BOARD TRAP
Disrupting this seal/attachment underlies some anti-resorptive strategies; intact sealing is required for acidification to work.

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3. Ruffled border — H+ pump

WHAT YOU SEE
Folded membrane with acid

WHAT IT ENCODES
The ruffled border houses a vacuolar H+-ATPase proton pump that acidifies the lacuna to pH ~4.5, dissolving the mineral (hydroxyapatite) phase. Acid solubilizes mineral; enzymes then digest matrix.

BOARD TRAP
Osteopetrosis = defective osteoclast acidification (carbonic anhydrase II or H+-ATPase/CLCN7 mutation) → dense brittle bone, marrow failure, anemia — bone that is too DENSE, not too thin.

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4. Cathepsin K — degrades collagen

WHAT YOU SEE
Saw cutting collagen

WHAT IT ENCODES
After acid dissolves mineral, cathepsin K (active at low pH) degrades the type I collagen matrix. CTX (C-telopeptide) released from this is the serum resorption marker.

BOARD TRAP
Cathepsin K loss-of-function = pycnodysostosis (Toulouse-Lautrec). Odanacatib (cathepsin K inhibitor) was dropped for stroke risk — distinct from denosumab.

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5. Carbonic anhydrase II — H+

WHAT YOU SEE
CA II enzyme generating protons

WHAT IT ENCODES
Carbonic anhydrase II generates the H+ that the ruffled-border pump exports. Essential for acidifying the resorption pit.

BOARD TRAP
CA II deficiency = osteopetrosis WITH renal tubular acidosis and cerebral calcification — a specific syndromic triad on the board.

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6. OB/OC/stromal present RANKL

WHAT YOU SEE
Orc holding RANKL flag

WHAT IT ENCODES
Osteoblasts and stromal cells display RANKL on their surface. RANKL is the central pro-resorptive signal — almost ALL osteotropic hormones (PTH, 1,25-D, etc.) act by raising osteoblast RANKL, NOT directly on osteoclasts.

BOARD TRAP
The one exception is CALCITONIN, which acts DIRECTLY on the osteoclast (calcitonin receptor) to inhibit it — every other major hormone routes through the osteoblast/RANKL.

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7. RANKL — fuse via M-CSF + RANK

WHAT YOU SEE
RANKL binding RANK on osteoclast

WHAT IT ENCODES
RANKL binds RANK on osteoclast precursors; with M-CSF it drives differentiation, fusion, and activation. This is THE master resorption switch.

BOARD TRAP
PTH/PTHrP, 1,25-D, IL-6, and glucocorticoids all increase RANKL — the final common pathway for hypercalcemia of malignancy and steroid-induced bone loss.

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8. OPG — soluble decoy → osteoclast OFF

WHAT YOU SEE
OPG decoy intercepting RANKL

WHAT IT ENCODES
Osteoprotegerin (OPG) is a soluble decoy receptor that binds RANKL and prevents RANK engagement → osteoclast OFF. The RANKL:OPG ratio sets net resorption.

BOARD TRAP
Estrogen RAISES OPG and lowers RANKL → estrogen loss (menopause) tips the ratio toward resorption — the mechanism of postmenopausal osteoporosis.

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9. Denosumab — anti-RANKL antibody

WHAT YOU SEE
DENOSUMAB labeled syringe/figure

WHAT IT ENCODES
Denosumab is a monoclonal anti-RANKL antibody (a pharmacologic OPG mimic) → potent anti-resorptive. Renally safe (usable in CKD, unlike bisphosphonates). Dosed q6 months SC.

BOARD TRAP
Stopping denosumab causes REBOUND vertebral fractures from a turnover surge — must transition to a bisphosphonate; do NOT simply discontinue. Also causes hypocalcemia if vit-D deplete.

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10. PTH — RANKL via OB

WHAT YOU SEE
Hormone-arch angel: PTH

WHAT IT ENCODES
PTH acts on osteoBLASTS (which carry the PTH1 receptor) to raise RANKL and lower OPG → indirect osteoclast activation; it also reduces renal Ca excretion and stimulates 1-alpha-hydroxylase.

BOARD TRAP
Osteoclasts have NO PTH receptor — PTH effect on resorption is entirely indirect via osteoblast RANKL.

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11. 1,25(OH)2D — RANKL via OB

WHAT YOU SEE
Hormone-arch figure: calcitriol

WHAT IT ENCODES
Active vitamin D (calcitriol) increases gut Ca/PO4 absorption and, via osteoblast RANKL, supports osteoclast maturation to mobilize Ca when needed.

BOARD TRAP
In CKD-MBD, low calcitriol drives secondary hyperPTH; over-replacing can cause hypercalcemia/vascular calcification — balance is the board point.

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12. Estrogen — ↓ osteoclast number

WHAT YOU SEE
Hormone-arch figure: estrogen

WHAT IT ENCODES
Estrogen suppresses osteoclastogenesis (raises OPG, lowers RANKL, promotes osteoclast apoptosis). Its loss accelerates resorption → menopausal bone loss.

BOARD TRAP
SERMs (raloxifene) mimic bone-protective estrogen but RAISE VTE risk and do not relieve hot flashes — distinct trade-off from estrogen replacement.

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13. Calcitonin — DIRECT on osteoclast

WHAT YOU SEE
Hormone-arch figure: calcitonin

WHAT IT ENCODES
Calcitonin (from thyroid C-cells) is the ONLY major hormone acting directly on the osteoclast calcitonin receptor to inhibit resorption. Weak/short-lived effect in humans; minor role in Ca homeostasis.

BOARD TRAP
Calcitonin is a tumor marker for MEDULLARY thyroid carcinoma — but thyroidectomy does NOT cause hypocalcemia from calcitonin loss (PTH dominates Ca control).

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14. Formation markers — P1NP/BSAP

WHAT YOU SEE
Ledger: FORMATION column

WHAT IT ENCODES
Bone FORMATION markers: P1NP (procollagen type I N-propeptide), bone-specific alkaline phosphatase (BSAP), osteocalcin. P1NP rises on teriparatide and tracks anabolic response.

BOARD TRAP
Don't mix these with resorption markers — P1NP up = building; CTX up = breaking down.

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15. Resorption markers — CTX/NTX

WHAT YOU SEE
Ledger: RESORPTION column

WHAT IT ENCODES
Bone RESORPTION markers: CTX and NTX (collagen telopeptides), pyridinoline. CTX falls with anti-resorptives and is used to confirm bisphosphonate/denosumab effect.

BOARD TRAP
A persistently elevated CTX on therapy suggests nonadherence or secondary cause; a suppressed CTX before dental surgery has been linked (debated) to ONJ risk.
